

Class 12th: Autonomous Agents & The Model Context Protocol (MCP)

Goal: Explore state-of-the-art architectures, building local tool connections, safe agent execution parameters.

Phase 1: Engineering Logic (Classes 1-5 — Refresher Classes)

Focus: Re-syncing students with prompt architecture, data schemas, and API communication logic.

Session	Topic	Core Learning Objectives & Details
Class 1	Complete Cloud Stack Review	Re-orienting students with LLM paradigms, token allocation, and JSON outputs.
Class 2	Chaining & API Architecture Quick-Build	Building a fast automated script pipeline from the Class 11 curriculum.
Class 3	Advanced Data Parsing	Reviewing how to map functional data attributes across model processing blocks.
Class 4	Introduction to the Agent Concept	From deterministic pipelines to dynamic decision-making loops (Reasoning + Acting = ReAct framework).
Class 5	Configuring Development Sandboxes	Setting up local project directory folders, script environments, and safe tool sandbox spaces.

Phase 2: MCP Integration Core (Classes 6-15)

Focus: Utilize open systems schema integrations (Model Context Protocol) to safely map engine logic directly to local storage layers and filesystems.

Session	Topic	Core Learning Objectives & Details
Class 6	Introduction to Model Context Protocol (MCP)	What is MCP? How does it bridge the gap between cloud LLMs and local development environments?
Class 7	MCP Architecture Anatomy	Understanding MCP Clients, MCP Hosts (like Claude Desktop), and MCP Servers.
Class 8	Setting Up Your First Local MCP Server	Connecting an AI safely to read and write to a specific folder on your local computer.
Class 9	Mapping the Filesystem Securely	Setting strict agent execution parameters so the AI can read data without breaking core operating systems.
Class 10	SQLite/Database MCP Integration	Letting an agent read, query, and modify local databases using conversational commands.
Class 11	Web Search MCP Connections	Giving local tools direct access to system terminal search utilities.
Class 12	Memory and State Management	Giving your agent a persistent local text file where it logs its long-term memory across sessions.
Class 13	Multi-tool Mapping	Combining filesystem tools, browser access tools, and mathematical computation engines into one unified system.
Class 14	Debugging MCP Schemas	Identifying connection drops and configuration mismatches between the host and the server.
Class 15	MCP Gateway Challenge	Successfully executing a complex query using three distinct local MCP tools.

Phase 3: Continuous Loop Code (Classes 16-25)

Focus: Design multi-stage feedback loops enabling processing blocks to parse, review, and adjust active operations autonomously.

Session	Topic	Core Learning Objectives & Details
Class 16	Building Autonomous Critique Systems	How an agent reviews its own work recursively until it hits a standard threshold.
Class 17	Infinite Loop Prevention (Guardrails)	Implementing termination counters (N_max_steps) and timeout metrics so loops don't spin endlessly.
Class 18	State Machine Logic	Designing agents that transition through explicit operational phases (e.g., Idle -> Researching -> Drafting -> Verifying -> Done).
Class 19	Handling Unpredicted Tools Failures	Training agents to try a different tool or strategy when the primary path errors out.
Class 20	Ingesting Active System Feedback	Allowing agents to read live terminal errors and write code modifications to correct themselves.
Class 21	Multi-Agent Choreography	Introduction to specialized agent squads (e.g., Creator Agent talks to an Editor Agent).
Class 22	Human-in-the-Loop Interventions	Designing a checkpoint where the autonomous loop stops and asks for human permission before making big moves.
Class 23	Event-Driven Background Scripts	Writing scripts that sit silently and trigger agent execution loops only when specific events occur.
Class 24	Logging Agent Thought Steps	Exporting detailed internal logs so users can audit how the agent made a decision.
Class 25	Stress-testing Autonomous Loops	Intentional environment breakage to test system resiliency.

Phase 4: Production Agent Run (Classes 26-35)

Focus: Deploy a continuous workspace agent program that monitors directory spaces and builds compiled tracking indices.

Session	Topic	Core Learning Objectives & Details
Class 26	Capstone Blueprinting	Architecture design for a complex, production-grade workspace supervisor agent.
Class 27	Initializing the Sandbox Workspace	Hardening local storage paths and preparing file directories.
Class 28	Activating the MCP Data Mesh	Binding the customized MCP protocols to the agent's core brain.
Class 29	Implementing the Multi-Stage Feedback Engine	Wiring autonomous review structures and validation checkers.
Class 30	Creating Compiled Tracking Indices	Programming the agent to maintain an updated master summary file of everything happening in the workspace.
Class 31	Human Verification Checkpoints	Implementing secure UI approvals for critical actions.
Class 32	Edge Case Optimization	Testing the system with massive files, duplicate records, and conflicting instructions.
Class 33	Performance Hardening	Minimizing code block overhead and refining token parameters for faster runs.
Class 34	Going Live: The Continuous Run	Deploying the background agent for its first long-duration automated monitoring task.
Class 35	Agent Exhibition Summit	Grand finale presentation of fully functional, autonomous cloud/local agent architectures.